



Hydrogen 1 H 1.00794		← 1s →																Helium 2 He 4.00260																																																																																															
3 Li 6.941 4 Be 9.0122		2s																2p						5 B 10.811 6 C 12.011 7 N 14.007 8 O 15.999 9 F 18.998 10 Ne 20.180																																																																																									
11 Na 22.990 12 Mg 24.305		3s																3d										3p						13 Al 26.982 14 Si 28.086 15 P 30.974 16 S 32.065 17 Cl 35.453 18 Ar 39.948																																																																															
19 K 39.098 20 Ca 40.078		4s		21 Sc 44.956 22 Ti 47.88 23 V 50.942 24 Cr 51.996 25 Mn 54.938 26 Fe 55.845 27 Co 58.933 28 Ni 58.693 29 Cu 63.546 30 Zn 65.38										4d		31 Ga 69.723 32 Ge 72.64 33 As 74.922 34 Se 78.96 35 Br 79.904 36 Kr 83.80						4p		37 Rb 85.468 38 Sr 87.62						5s		39 Y 88.906 40 Zr 91.224 41 Nb 92.906 42 Mo 95.94 43 Tc [98] 44 Ru 101.07 45 Rh 102.91 46 Pd 106.42 47 Ag 107.87 48 Cd 112.41										5d		49 In 114.82 50 Sn 118.71 51 Sb 121.76 52 Te 127.60 53 I 126.90 54 Xe 131.29						5p		55 Cs 132.91 56 Ba 137.33						6s		57-70 Lanthanide series 71 Lu 174.967 72 Hf 178.49 73 Ta 180.95 74 W 183.84 75 Re [186.21] 76 Os 190.23 77 Ir 192.22 78 Pt 195.08 79 Au 196.97 80 Hg 200.59										6d		81 Tl 204.38 82 Pb 207.2 83 Bi 208.98 84 Po [209] 85 At [210] 86 Rn [222]						7p		87 Fr [223] 88 Ra [226]						7s		89-102 Actinide series 103 Lr [260] 104 Rf [261] 105 Db [262] 106 Sg [266] 107 Bh [264] 108 Hs [277] 109 Mt [268] 110 Uun [271] 111 Uuu [272] 112 Uub [277]										7d		113 Nh [284] 114 Fl [289] 115 Mc [288] 116 Lv [293] 117 Ts [294] 118 Og [294]						7p		119 Uut [285] 120 Uuq [289]					

* Lanthanide series

57 La 138.91	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04
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** Actinide series

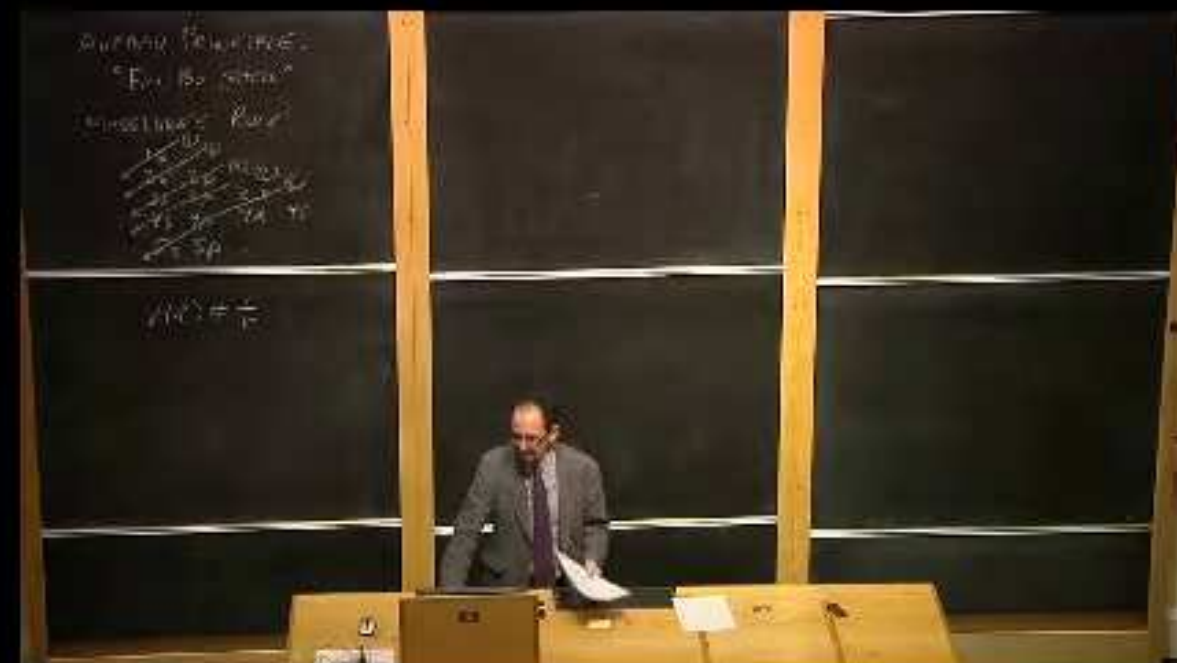
89 Ac [227]	90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np 237.0	94 Pu [244]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]
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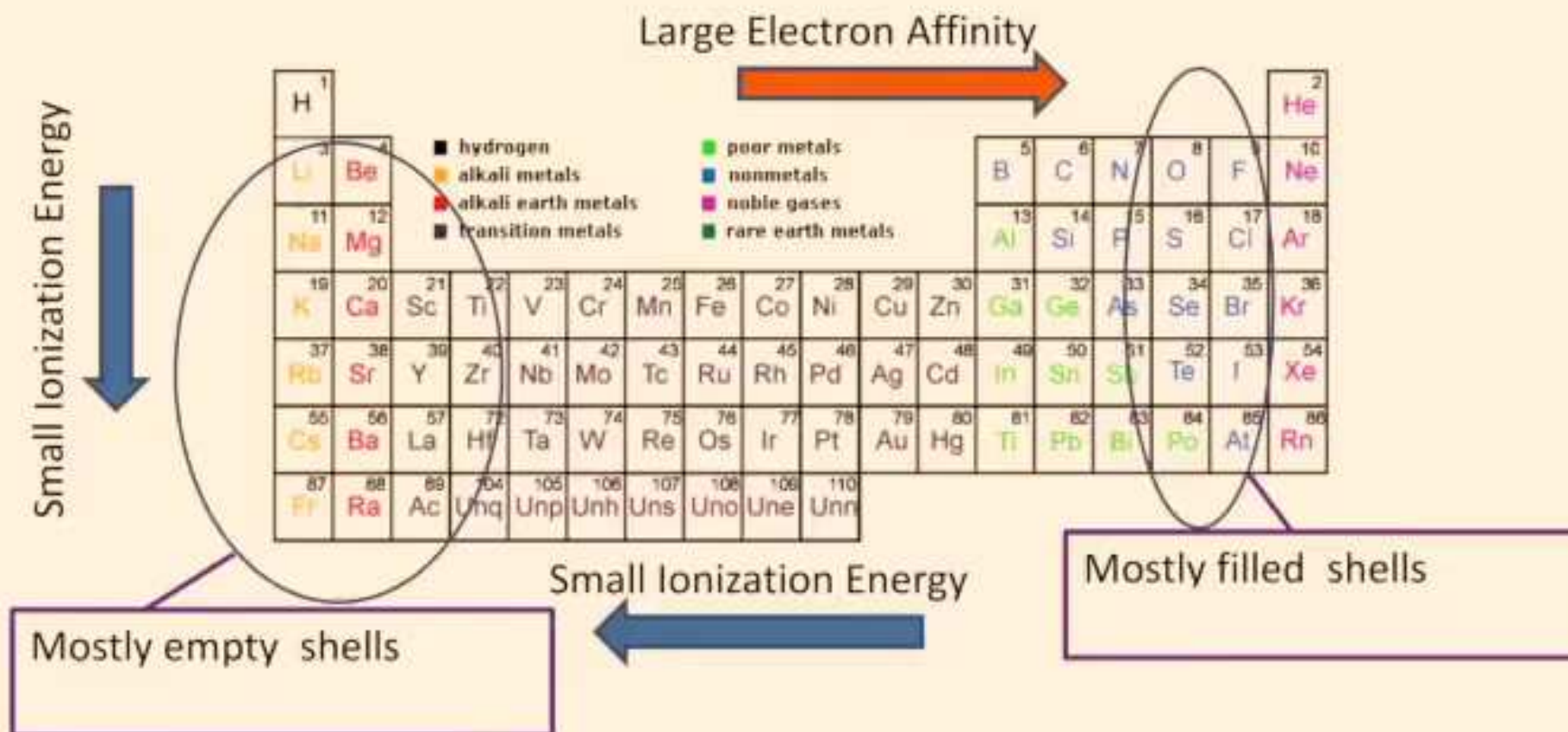
4f



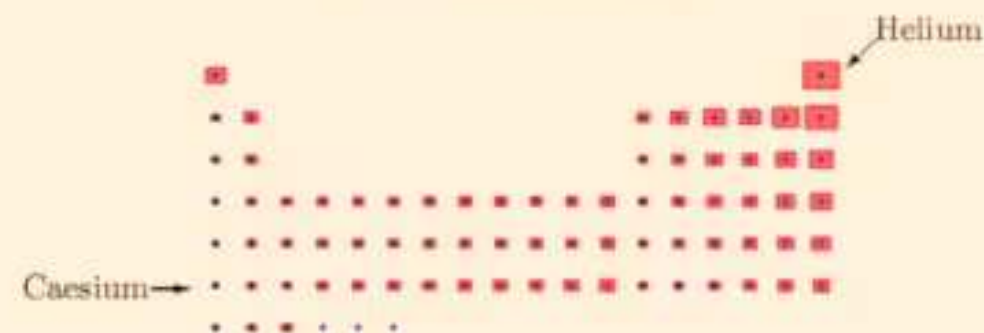
Type of Bonding	Description	Typical of which compounds	Typical Properties
Ionic	Electron is transferred from one atom to another, and the resulting ions attract each other	Binary compounds made of constituents with very different electronegativity: Ex. group 1-7 such as NaCl or group 2-7 compounds.	• Hard, Very Brittle • High Melting Temperature • Electrical Insulator • Water Soluble
Covalent	Electron is shared equally between two atoms forming a bond. Energy lowered by delocalization of wavefunction	Compounds made of constituents with similar electronegativities (ex. 3-5 compounds such as GaAs), or solids made of one element only such as Diamond (C)	• Very Hard (Brittle) • High Melting Temperature • Electrical Insulators or Semiconductors
Metallic Bonds	Electrons delocalized throughout the solid forming a glue between positive ions.	Metals. Left and Middle of Periodic Table.	• Ductile, Malleable (due to non-directional nature of bond. Can be hardened by preventing dislocation motion with impurities) • Lower Melting Temperature • Good electrical and thermal conductors.
Molecular (van der Waals or Fluctuating Dipole)	No transfer of electrons. Dipole moments on constituents align to cause attraction. Bonding strength increases with size of molecule or polarity of constituent.	Noble Gas Solids, Solids made of Non-Polar (or slightly polar) Molecules Binding to Each Other (Wax)	• Soft, Weak • Low Melting Temperature • Electrical Insulators
Hydrogen	Involves Hydrogen ion bound to one atom but still attracted to another. Special case because H is so small.	Important in organic and biological materials	• Weak Bond (stronger than VdW though) • Important for maintaining shape of DNA and proteins

Table 4.1: Types of Bonds in Solids. This table should be thought of as providing rough rules. Many materials show characteristics intermediate between two (or more!) classes.





First Ionization Energies



First Electron Affinities

